# Validation Architecture in delaunay

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**Author TODO.** Write the abstract in your own words.

Additional Key Words and Phrases: TODO author-supplied keywords

## 1 INTRODUCTION

- **Author TODO.** State the problem and motivation.
- **Author TODO.** Explain why validation architecture matters for a scientific computational-geometry library.
- **Author TODO.** Summarize the central contribution in your own words.
- **Author TODO.** Preview the paper structure.

## 2 BACKGROUND AND DEFINITIONS

- **Author TODO.** Define the triangulation, simplex, ridge, facet, and realization terms used throughout the paper.
- **Author TODO.** Introduce PL-manifold terminology and the link condition.
- **Author TODO.** Introduce Euclidean, toroidal, and spherical realization models.
- **Author TODO.** Introduce Delaunay and future geometric-predicate terminology.

## 3 VALIDATION HIERARCHY

### 3.1 Level 1: Element Validity

- **Author TODO.** Describe what individual vertices, simplices, facets, and coordinates must prove locally.
- **Author TODO.** Explain what this level deliberately does not check.

### 3.2 Level 2: Combinatorial Consistency

- **Author TODO.** Describe adjacency, incidence, neighbor symmetry, and TDS integrity.
- **Author TODO.** Explain the separation from topology and geometry.

### 3.3 Level 3: Intrinsic PL Topology

- **Author TODO.** State the PL-manifold and boundary conditions.
- **Author TODO.** Discuss pseudomanifold and strict PL-manifold policy choices.
- **Author TODO.** Explain why this layer is independent of Euclidean, toroidal, and spherical realizations.

### 3.4 Level 4: Valid Realization

- **Author TODO.** Use the terminology stack: Level 4 Valid Realization; Euclidean/toroidal valid affine-chart realization; spherical valid spherical realization; Level 5 Geometric Predicate Satisfaction / Delaunay Optimality.
- **Author TODO.** Describe valid realization in the active ambient model.
- **Author TODO.** Compare Euclidean/toroidal affine-chart realizations and spherical realizations.
- **Author TODO.** Explain why realization validity is a precondition for geometric predicates.

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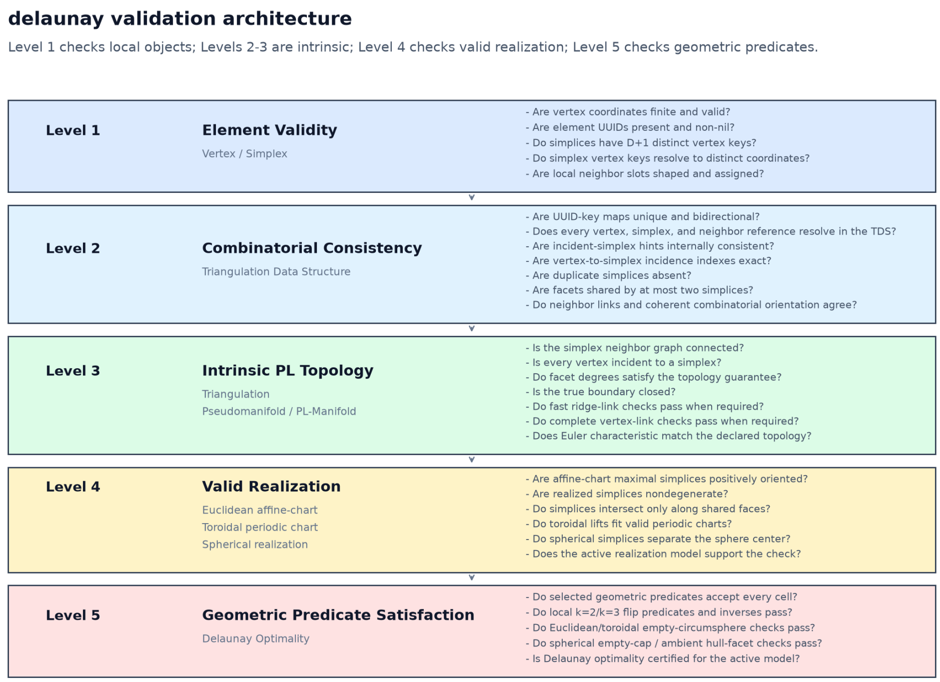


Fig. 1. Overview of the validation hierarchy in delaunay

### 3.5 Level 5: Geometric Predicates

- **Author TODO.** Describe Delaunay as the first implemented predicate family.
- **Author TODO.** Explain how Euclidean, toroidal, and spherical Delaunay predicates differ.
- **Author TODO.** Leave room for regular, weighted, constrained, Gabriel, alpha, and future predicate families.

## 4 IMPLEMENTATION ARCHITECTURE

- **Author TODO.** Map the five validation levels onto the Rust API surface.
- **Author TODO.** Describe the role of fast-fail checks, diagnostics, reports, and cumulative validation.
- **Author TODO.** Explain construction-time validation policy choices such as `ExplicitOnly` and `OnSuspicion`.
- **Author TODO.** Describe how spherical topology fits as a coordinate/metric backend rather than a new simplex dimension.

## 5 DIAGNOSTICS AND REPRODUCIBLE FIGURES

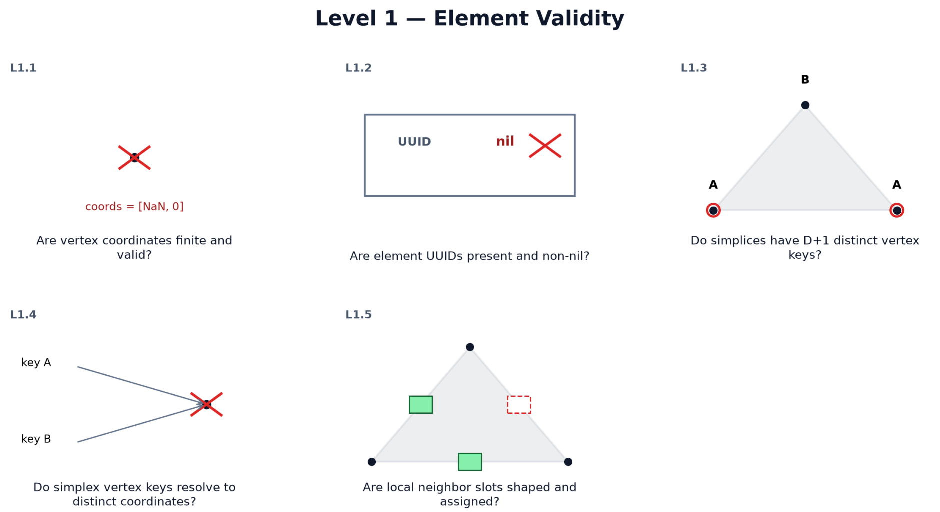


Fig. 2. **Author TODO.** Describe the Level 1 validation witness figure.

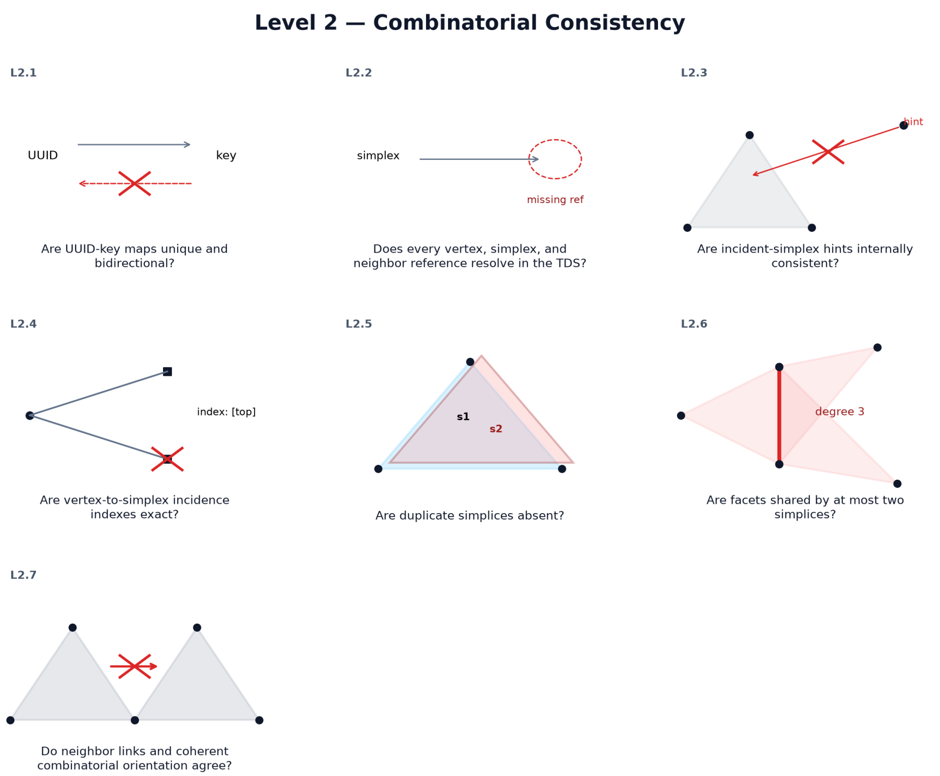


Fig. 3. **Author TODO.** Describe the Level 2 validation witness figure.

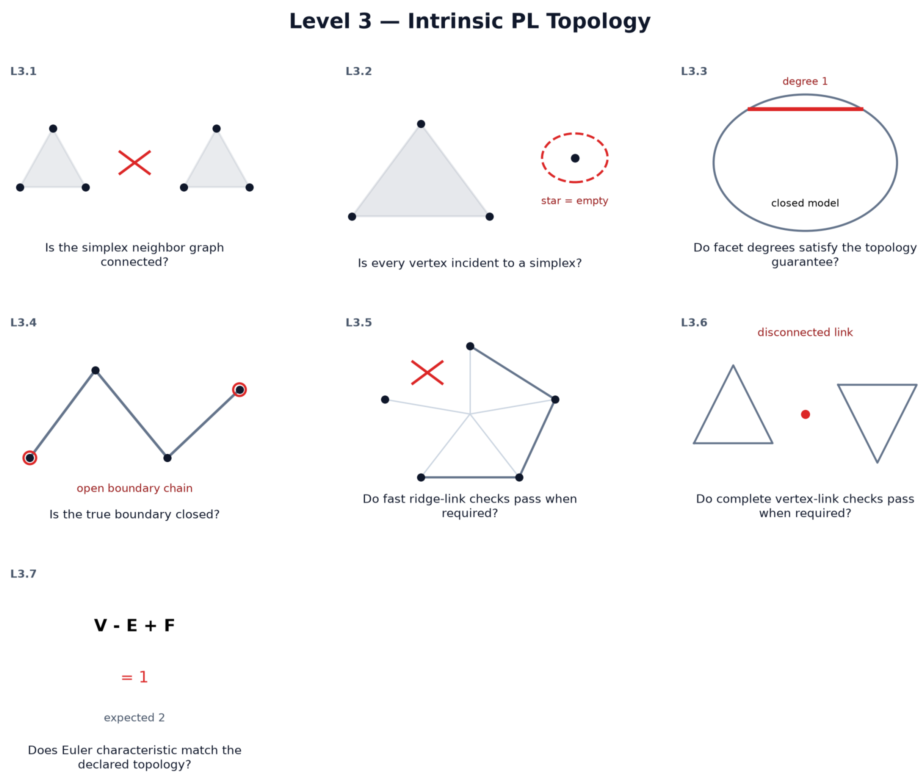


Fig. 4. **Author TODO.** Describe the Level 3 validation witness figure.

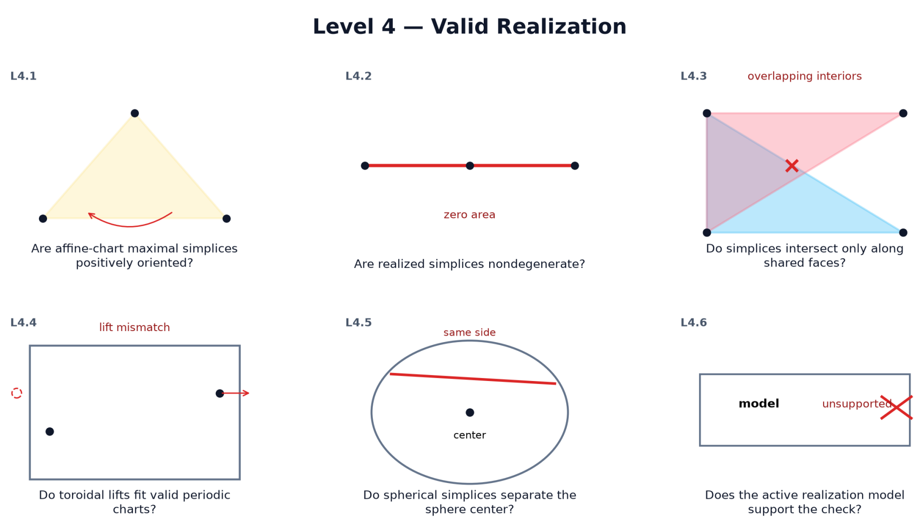


Fig. 5. **Author TODO.** Describe the Level 4 validation witness figure.

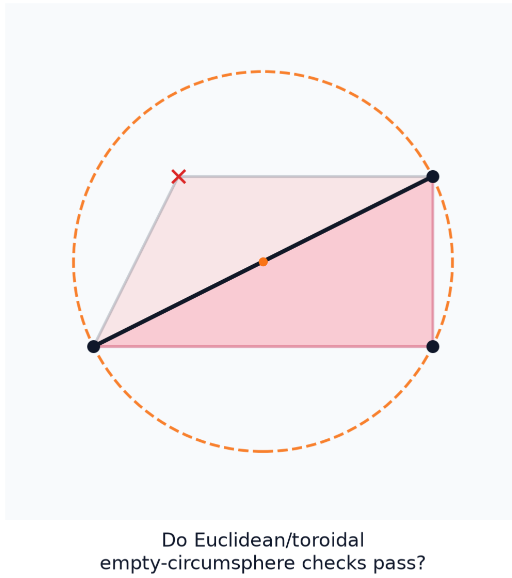


Fig. 6. **Author TODO.** Describe the Level 5 validation witness figure.

- **Author TODO.** Explain how the validation notebook generates paper figures.
- **Author TODO.** Describe representative diagnostics without duplicating API documentation.
- **Author TODO.** Connect figure generation to the reproducibility story.

## 6 RELATED WORK

- **Author TODO.** Place PL topology references and Pachner/Lickorish results.
- **Author TODO.** Place Euclidean robust-predicate and Delaunay references.
- **Author TODO.** Place spherical Delaunay and Riemannian-manifold references.
- **Author TODO.** Clarify which claims are architectural observations rather than literature claims.

## 7 DISCUSSION AND FUTURE WORK

- **Author TODO.** Discuss extensibility to additional realization models and predicate families.
- **Author TODO.** Discuss limitations of the current implementation.
- **Author TODO.** Identify open validation and benchmarking questions.

## 8 CONCLUSION

- **Author TODO.** Write the conclusion in your own words.

## ACKNOWLEDGMENTS

The author thanks Steven Carlip for discussions and guidance over the years on Causal Dynamical Triangulation, which motivated this work, and for encouragement and questions on this library, which motivated this paper.

## REFERENCES


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